

Spring 1978

Problem 1 (Sp78). Let $k \geq 0$ be an integer and define a sequence of maps

$$f_n : \mathbb{R} \rightarrow \mathbb{R} \quad f_n(x) = \frac{x^k}{x^2 + n} \quad n = 1, 2, \dots$$

For which values of k does the sequence converge uniformly on $\mathbb{R}$? On every bounded subset of $\mathbb{R}$?

Problem 2 (Sp78). Prove that a map $f : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is continuous only if its graph G_f is closed in $\mathbb{R}^n \times \mathbb{R}^n$. Is the converse true?

Note: See also ????

Problem 3 (Sp78). Let $f : \mathbb{C} \rightarrow \mathbb{C}$ be a nonconstant entire function. Prove that $f(\mathbb{C})$ is dense in $\mathbb{C}$.

Problem 4 (Sp78, Sp83, Sp97). Evaluate $\int_{-\infty}^{\infty} \frac{\sin^2 x}{x^2} dx$.

Problem 5 (Sp78). Let $\mathbb{Z}_n$ denote the ring of integers modulo n . Let $\mathbb{Z}_n[x]$ be the ring of polynomials with coefficients in $\mathbb{Z}_n$. Let $\mathfrak{I}$ denote the ideal in $\mathbb{Z}_n[x]$ generated by $x^2 + x + 1$.

1. For which values of n , $1 \leq n \leq 10$, is the quotient ring $\mathbb{Z}_n[x]/\mathfrak{I}$ a field?
2. Give the multiplication table for $\mathbb{Z}_2[x]/\mathfrak{I}$.

Problem 6 (Sp77, Sp78). Let $\mathbb{F} \subset \mathbb{K}$ be fields, and a and b elements of $\mathbb{K}$ which are algebraic over $\mathbb{F}$. Show that $a + b$ is algebraic over $\mathbb{F}$.

Problem 7 (Sp78). What is the volume enclosed by the ellipsoid

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1?$$

Problem 8 (Sp78). Consider the differential equation $\frac{dx}{dt} = x^2 + t^2$ with initial condition $x(0) = 1$.

1. Prove that for some $b > 0$, there is a solution defined for $t \in [0, b]$.
2. Find an explicit value of b having the property in Part 1.

3. Find a $c > 0$ such that there is no solution on $[0, c]$.

Problem 9 (Sp78, Sp82, Su82, Fa90). Determine the Jordan Canonical Form of the matrix

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 0 & 4 & 5 \\ 0 & 0 & 4 \end{pmatrix}$$

Problem 10 (Sp78). Suppose A is a real $n \times n$ matrix.

1. Is it true that A must commute with its transpose?
2. Suppose the columns of A (considered as vectors) form an orthonormal set; is it true that the rows of A must also form an orthonormal set?

Problem 11 (Sp78, Sp18, Fa21). Show that there is a complex analytic function defined on the set $U = \{z \in \mathbb{C} \mid |z| > 4\}$ whose derivative is

$$f(z) = \frac{z}{(z-1)(z-2)(z-3)}$$

Is there a complex analytic function on U whose derivative is

$$g(z) = \frac{z^3}{(z-1)(z-2)(z-3)} ?$$

Problem 12 (Sp78). Prove that the uniform limit of a sequence of complex analytic functions is complex analytic. Is the analogous theorem true for real analytic functions?

Problem 13 (Sp78, Fa10). 1. For which real numbers $\alpha > 0$ does the differential equation

$$\frac{dx}{dt} = x^\alpha \quad x(0) = 0$$

have a solution on some interval $[0, b]$, $b > 0$?

2. For which values of α are there intervals on which two solutions are defined?

Problem 14 (Sp78, Fa81). Let G be a group of order 10 which has a normal subgroup of order 2. Prove that G is abelian.

Problem 15 (Sp78). Is $x^4 + 1$ irreducible over the field of real numbers? The field of rational numbers? A field with 16 elements?

Problem 16 (Sp78, Sp98). Let A and B denote real $n \times n$ symmetric matrices such that $AB = BA$. Prove that A and B have a common eigenvector in $\mathbb{R}^n$.

Problem 17 (Sp78). Evaluate $\iint_{\mathcal{A}} e^{-x^2-y^2} dx dy$ where

$$\mathcal{A} = \{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 \leq 1\}$$

Problem 18 (Sp78, Fa82, Fa86). Let M be a matrix with entries in a field $\mathbb{F}$. The *row rank* of M over $\mathbb{F}$ is the maximal number of rows which are linearly independent (as vectors) over $\mathbb{F}$. The *column rank* is similarly defined using columns instead of rows.

1. Prove row rank = column rank.
2. Find a maximal linearly independent set of columns of

$$\begin{pmatrix} 1 & 0 & 3 & -2 \\ 2 & 1 & 2 & 0 \\ 0 & 1 & -4 & 4 \\ 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \end{pmatrix}$$

taking $\mathbb{F} = \mathbb{R}$.

3. If $\mathbb{F}$ is a subfield of $\mathbb{K}$, and M has entries in $\mathbb{F}$, how is the row rank of M over $\mathbb{F}$ related to the row rank of M over $\mathbb{K}$?

Problem 19 (Sp78). Let $f : [0, 1] \rightarrow \mathbb{R}$ be Riemann integrable over $[b, 1]$ for all b such that $0 < b \leq 1$.

1. If f is bounded, prove that f is Riemann integrable over $[0, 1]$.
2. What if f is not bounded?

Problem 20 (Sp78). Consider the system of equations

$$3x + y - z + u^4 = 0$$

$$x - y + 2z + u = 0$$

$$2x + 2y - 3z + 2u = 0$$

1. Prove that for some $\varepsilon > 0$, the system can be solved for (x, y, u) as a function of $z \in [-\varepsilon, \varepsilon]$, with $x(0) = y(0) = u(0) = 0$. Are such functions $x(z)$, $y(z)$ and $u(z)$ continuous? Differentiable? Unique?
2. Show that the system cannot be solved for (x, y, z) as a function of $u \in [-\delta, \delta]$, for all $\delta > 0$.